

CRITICAL

Account Takeover

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Routed to: Compliance Team · Fraud Risk Team · Tm Analytics Team · Policy Controls Team

EXECUTIVE SUMMARY

Russian-origin threat actors are executing account takeovers by phishing for Signal Backup Recovery Keys. This threat primarily targets UOB Privilege Banking and Business Banking clients, with a potential worst-case financial exposure of SGD 15.1 million. Immediate action is required to deploy a targeted customer advisory and brief front-line staff to mitigate imminent risk.

THREAT INTELLIGENCE BRIEF

A novel account takeover typology has emerged where threat actors use sophisticated social engineering and phishing tactics to deceive victims into disclosing their 30-digit Signal Backup Recovery Key. The attack unfolds in stages: 1) The victim receives a phishing message (SMS or email) creating a false sense of urgency, such as a fake security alert. 2) The victim is directed to a malicious website impersonating a legitimate Signal or security service. 3) The victim is manipulated into entering their recovery key. 4) The attacker uses this key to restore the victim's entire Signal message history onto a new device, gaining persistent, undetected access to their account. This allows the attacker to impersonate the victim, intercept sensitive communications, and gather intelligence for subsequent, high-value payment or investment fraud.

UOB RELEVANCE ASSESSMENT

Gate	Result	Details
Gate 1 — Novelty	✓ PASSED	No existing TM rule covers this typology; No similar signal alerted in last 90 days; Moderate novelty score (0.50)
Gate 2 — Customer Exposure	✓ PASSED	✓ MULTIPLE customers in RU; ~1,000 at risk
Gate 3 — Control Gap	✓ PASSED	✓ 100% of attacks undetected by existing TM rules; auth vulnerability: YES

Composite Risk Score: 80/100 — CRITICAL Priority

This threat is rated CRITICAL for UOB with a Composite Risk Score of 80.0/100. The attack vector is highly relevant as our high-net-worth segments, including UOB Privilege Banking and Business Banking clients, increasingly use encrypted messaging platforms like Signal for sensitive business and personal communication. A full gap analysis confirms this is a novel threat (Gate 1: PASSED), with an estimated 1,000 customers directly exposed (Gate 2: PASSED), and a 100% control gap as the attack occurs out-of-band, rendering existing monitoring ineffective (Gate 3: PASSED).

FINANCIAL IMPACT SIMULATION

Scenario	Exposure (SGD)
Baseline (no intervention)	SGD 980K

With current TM controls	SGD 980K
With proposed interventions	SGD 431K
P10 (optimistic)	SGD 1.5M
P50 (median)	SGD 4.7M
P90 (worst case)	SGD 15.1M

The potential financial impact is significant. The baseline expected loss from this typology is SGD 980,000. Probabilistic modeling indicates a 10th percentile loss (P10) of SGD 1.5 million and a 90th percentile worst-case scenario (P90) of SGD 15.1 million. Current controls are 100% ineffective against this out-of-band threat. However, simulations show that the implementation of proposed interventions can reduce the baseline loss by 56% to SGD 431,000.

RECOMMENDED ACTIONS

■ Immediate (<24 hours)

- Issue a targeted security advisory via email and UOB TMRW app notification to all Privilege Banking and Business Banking clients, detailing the Signal phishing threat and advising them never to share their recovery key.
- Distribute an internal threat bulletin to all relationship managers and front-line staff, outlining the attack methodology and key indicators of a compromised client.

■ Short-term (1–7 days)

- Develop and deploy a new TM_DETECTION rule to identify downstream payment fraud originating from this compromise, focusing on unusual high-value transfers to new payees from the at-risk cohort.
- Implement a targeted CUSTOMER_FRICTION step-up authentication challenge for any changes to contact details or for high-value transfers to new beneficiaries initiated by the identified at-risk customer segments.
- Launch a broader digital security campaign educating all customers on the risks of sharing credentials for third-party communication apps.

■ Strategic (1–4 weeks)

- Initiate a POLICY_CHANGE review to formalize guidelines for relationship managers on using third-party messaging apps for client communication, emphasizing approved channels and topics.
- Share anonymized indicators of compromise (IOCs) and threat actor tactics, techniques, and procedures (TTPs) with the FS-ISAC community to enhance sector-wide resilience.

PROPOSED INTERVENTION RULES

1. TM_DETECTION Risk: MEDIUM · Impact: ~25% · Auto-deployable: No

Detect suspicious high-value transfers to new beneficiaries, especially when initiated from an unrecognised device and authorized by SMS OTP, potentially indicating a post-Signal account takeover fraud attempt.

```
IF (Transaction_Amount > S$5,000 AND Beneficiary_Is_New_180_Days AND
Device_IP_Unrecognised_90_Days AND Auth_Method = 'SMS_OTP' AND
Customer_Contact_Details_Updated_7_Days) THEN Flag_For_Manual_Review_And_Hold_Transaction
```

■ Requires Head of Fraud Risk sign-off

2. CUSTOMER_FRICTION Risk: MEDIUM · Impact: ~30% · Auto-deployable: No

Introduce an additional authentication challenge for high-risk actions like changing registered mobile numbers or adding new high-value beneficiaries, especially if initiated from an unrecognised device.

```
IF (Action = 'Update_Registered_Mobile_Number' OR (Action = 'Add_New_Beneficiary' AND
Beneficiary_Transfer_Limit_Proposed > S$2,000)) AND Device_IP_Unrecognised_90_Days THEN
Require_UOB_Mighty_Push_Approval_OR_Call_Back_To_Pre_Registered_Number
```

■ Requires Head of Fraud Risk sign-off

3. POLICY_CHANGE Risk: HIGH · Impact: ~35% · Auto-deployable: No

Implement a mandatory cooling-off period for critical account changes to mitigate the risk of attackers immediately leveraging compromised Signal accounts to change contact details or add beneficiaries.

```
Implement a mandatory 12-hour cooling-off period for any successful update to a customer's
registered mobile number or activation of a new beneficiary for transfers exceeding S$5,000.
During this period, SMS OTPs will be sent to the previous mobile number, and transactions to
newly added beneficiaries will be capped at S$500.
```

■ Requires Head of Fraud Risk sign-off

4. ADVISORY Risk: LOW · Impact: ~15% · Auto-deployable: Yes

Issue a multi-channel customer advisory warning about phishing scams targeting Signal Backup Recovery Keys and other personal credentials, reinforcing safe digital practices.

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Disseminate advisory via UOB Mighty app, email, and website banner to all digital banking
customers. Content to include warnings against sharing Signal Backup Recovery Keys, verifying
unsolicited requests, and UOB's policy of never asking for such credentials.
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EVIDENCE SOURCES

- 1. Internal Threat Signal [ID: SIG-2023-4812] detailing phishing vector and threat actor TTPs.
- 2. Geospatial intelligence report linking campaign infrastructure to Russian-based threat actors.
- 3. Financial impact model [FIM-2023-11-02] based on a cohort of 1,000 exposed UOB customers.

ANALYST NOTES & CAVEATS

This attack vector bypasses traditional on-platform security controls by compromising the customer's communication channel directly. Financial fraud is a downstream effect. This alert is AI-generated. It must be reviewed by a qualified fraud risk analyst before operational action is taken.